

Git, Version Control & CI/CD

Project Documentation - Mini ETL Script

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Repository	github.com/bencybenedict/Git-Version-Control-CI-CD
Tools	Git, GitHub, GitHub Actions, Python 3.13, pytest, VS Code, PowerShell
Projects	1.1 Version a Mini ETL Script; 1.2 Branch-and-Merge a New Feature; 1.3 GitHub Repo with a Test Workflow

1. Overview

This project practises the complete Git workflow on a small ETL (Extract, Transform, Load) Python script: creating a repository, committing in logical steps, tagging a release, branching and resolving a merge conflict, and finally pushing to GitHub with an automated test workflow (CI) that runs pytest on every push.

2. Project 1.1 - Version a Mini ETL Script

Goal: practise the full Git loop on a real data script.

- Created `extract.py` and initialised the repo with `git init`.
- Added a `.gitignore` to ignore `__pycache__`, `.env` and `*.csv`.
- Made meaningful commits as the script grew: `extract`, `transform`, `load` functions.
- Tagged the finished version as `v1.0.0`.

```
900d9a9 feat: add extract function
9c316e2 feat: add transform function
3515061 feat: add load function (tag: v1.0.0)
```

Learned: `init`, `add`, `commit`, `.gitignore`, `tags`.

3. Project 1.2 - Branch-and-Merge a New Feature

Goal: simulate team development with a branch and a merge conflict.

- Created `feature/add-null-check` from `main` and added null-validation code (commits *add null validation* and *add ETL completion message*).
- Edited the same lines on `main` (*add processing message*) so that both branches changed the same part of `extract.py`.
- Ran `git merge feature/add-null-check`, which reported a content conflict. Resolved the conflict markers manually and committed the merge (*merge: resolve validation conflict*).

```

Windows PowerShell
PS C:\Users\HP\OneDrive\文档\Git Project> git add extract.py
PS C:\Users\HP\OneDrive\文档\Git Project> git commit -m "feat: add null validation"
[feature/add-null-check 51003bd] feat: add null validation
1 file changed, 11 insertions(+)
PS C:\Users\HP\OneDrive\文档\Git Project> git add extract.py
PS C:\Users\HP\OneDrive\文档\Git Project> git commit -m "feat: add ETL completion message"
[feature/add-null-check 9228ef2] feat: add ETL completion message
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\HP\OneDrive\文档\Git Project> git merge feature/add-null-check
Already up to date.
PS C:\Users\HP\OneDrive\文档\Git Project> git branch
* feature/add-null-check
  main
PS C:\Users\HP\OneDrive\文档\Git Project> git log --oneline --all --graph
* 9228ef2 (HEAD -> feature/add-null-check) feat: add ETL completion message
* 51003bd feat: add null validation
* 3515061 (tag: v1.0.0, main) feat: add load function
* 9c316e2 feat: add transform function
* 900d9a9 feat: add extract function
PS C:\Users\HP\OneDrive\文档\Git Project> git status
On branch feature/add-null-check
nothing to commit, working tree clean
PS C:\Users\HP\OneDrive\文档\Git Project> git switch main
Switched to branch 'main'
PS C:\Users\HP\OneDrive\文档\Git Project> git log --oneline
3515061 (HEAD -> main, tag: v1.0.0) feat: add load function
9c316e2 feat: add transform function
900d9a9 feat: add extract function
PS C:\Users\HP\OneDrive\文档\Git Project> git add extract.py
PS C:\Users\HP\OneDrive\文档\Git Project> git commit -m "feat: add processing message"
[main 7b6d373] feat: add processing message
1 file changed, 1 insertion(+)
PS C:\Users\HP\OneDrive\文档\Git Project> git merge feature/add-null-check
Auto-merging extract.py
CONFLICT (content): Merge conflict in extract.py

```

Figure 1 - Terminal: commits on the feature branch, new commit on main, and the merge conflict in extract.py

Learned: branching, merging, conflict resolution.

4. Project 1.3 - GitHub Repo with a Test Workflow

Goal: first taste of Continuous Integration.

4.1 Setup

- Pushed the repository to GitHub: `git push --set-upstream origin main`.
- Wrote a pytest test, `test_transform` in `test_extract.py`, which feeds string data to `transform()` and checks the output types and values.
- Added `.github/workflows/ci.yml` (workflow name *Python CI*) that runs pytest automatically.

4.2 Run #1 - workflow passes

After committing *ci: add GitHub Actions workflow* and *fix: update GitHub Actions workflow*, the first successful run completed in 15 seconds.

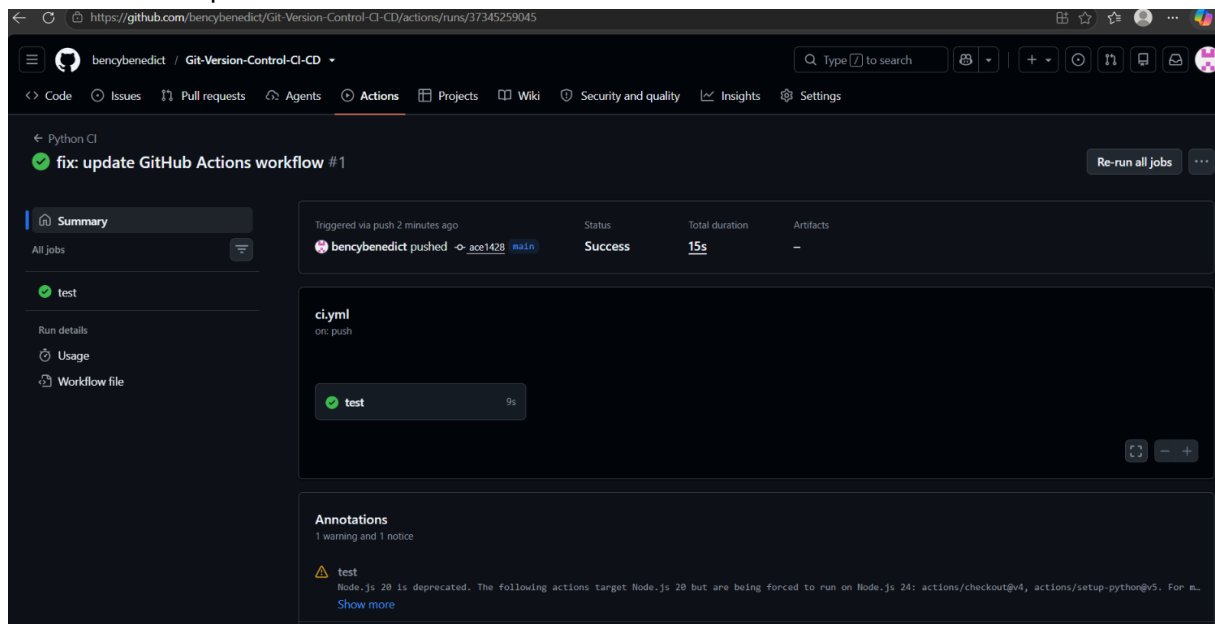


Figure 2 - GitHub Actions run #1: Success

4.3 Run #2 - intentionally broken test

To watch CI catch a failure, the assertion was deliberately changed to expect a quantity of 100 while the real result is 2.

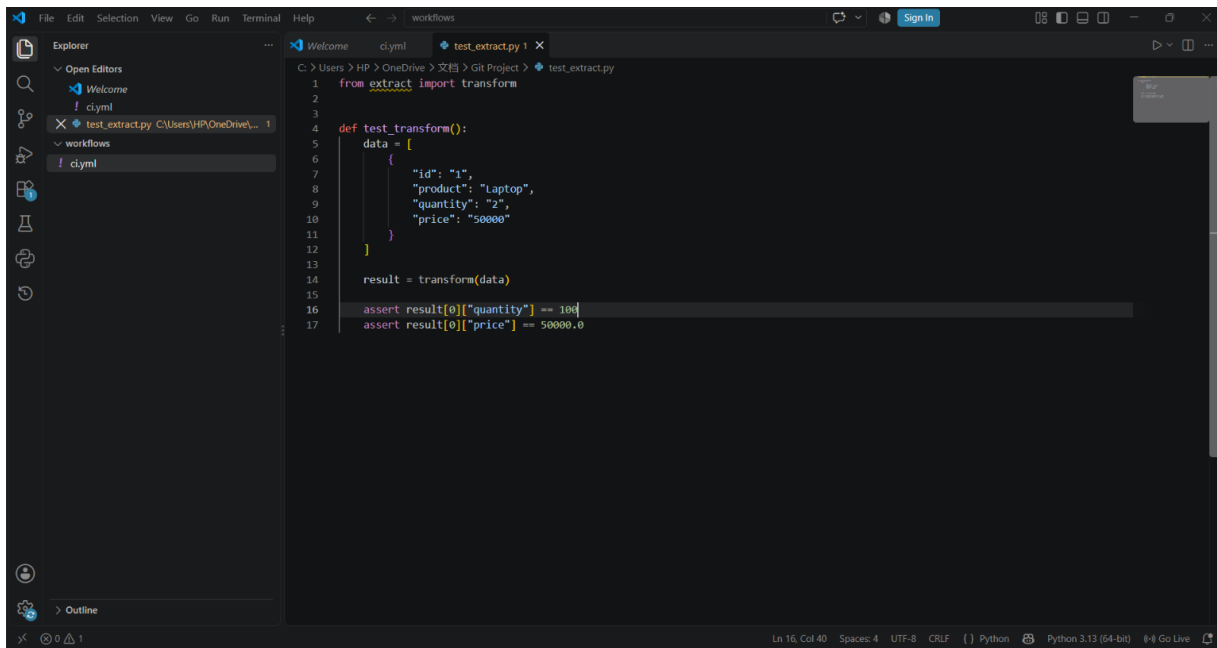


Figure 3 - VS Code: assertion changed to == 100 (broken on purpose)

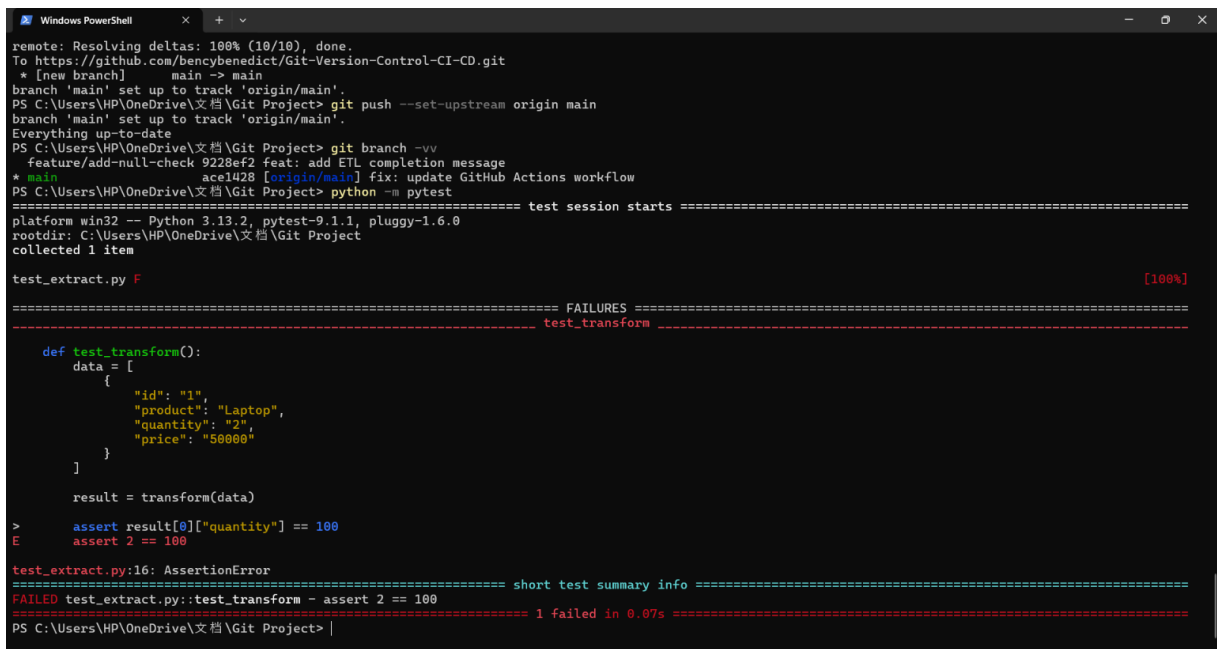


Figure 4 - Local pytest run: FAILED, assert 2 == 100

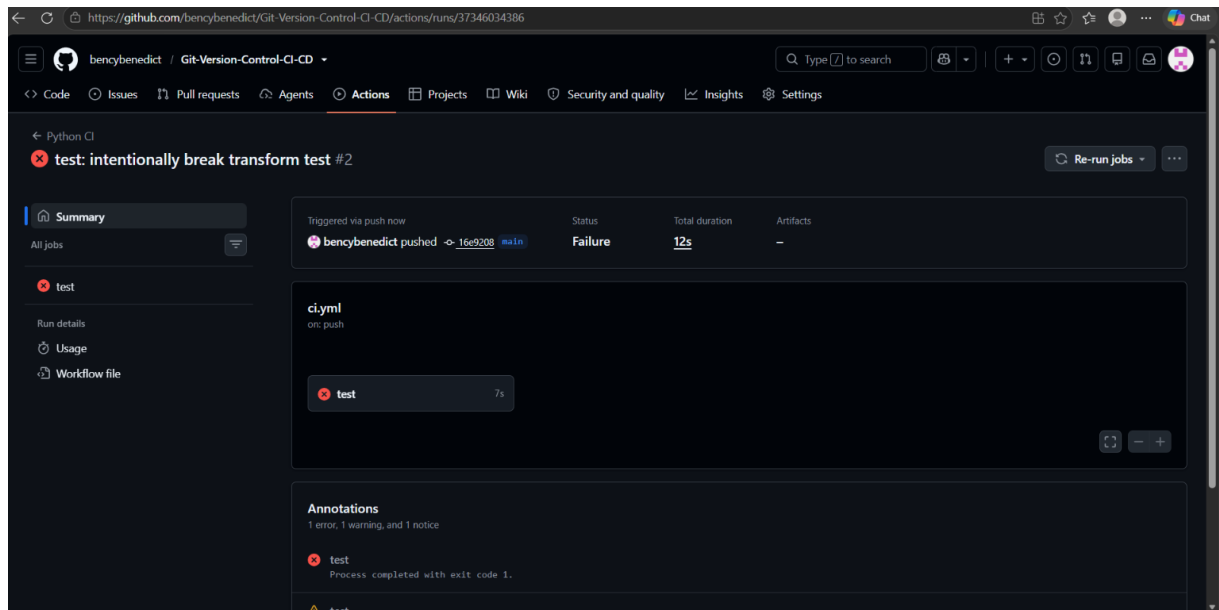


Figure 5 - GitHub Actions run #2: Failure (exit code 1)

4.4 Run #3 - fix and pass

The assertion was corrected back to the expected value 2, committed as *fix: correct transform test* and pushed. CI went green again.

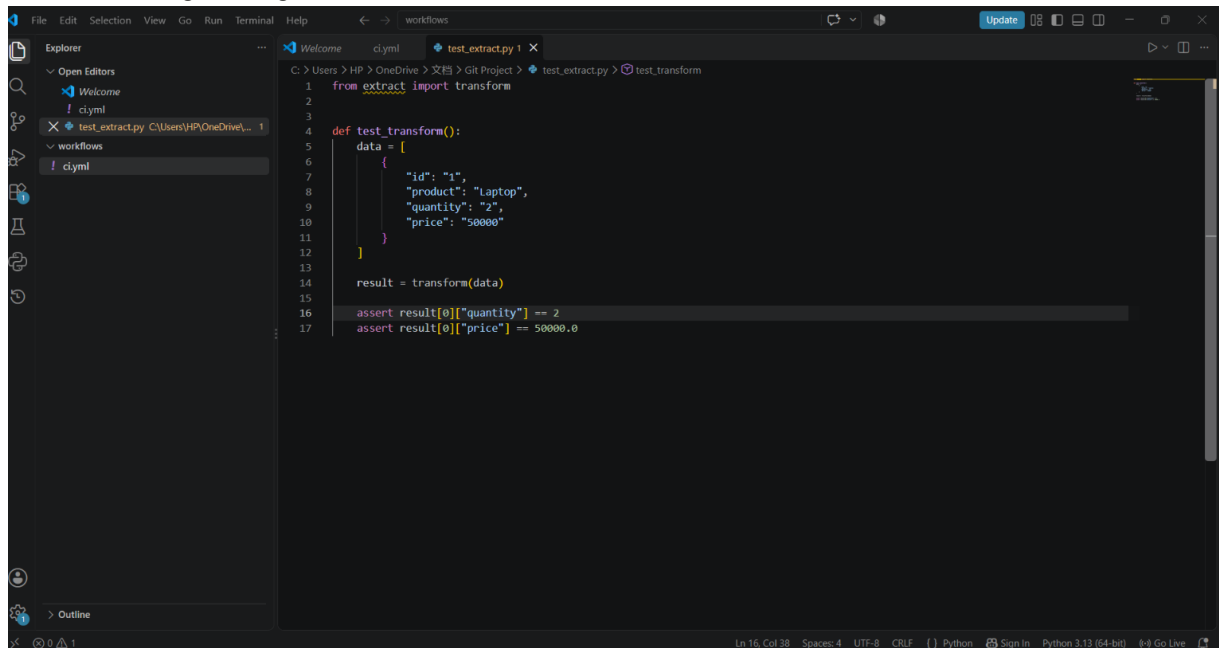


Figure 6 - VS Code: assertion corrected to == 2

```
Windows PowerShell
PS C:\Users\HP\OneDrive\文档\Git Project> git add test_extract.py
PS C:\Users\HP\OneDrive\文档\Git Project> git commit -m "fix: correct transform test"
[main c6b2738] fix: correct transform test
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\HP\OneDrive\文档\Git Project> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 304 bytes | 304.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/bencybenedict/Git-Version-Control-CI-CD.git
16e9208..c6b2738 main -> main
PS C:\Users\HP\OneDrive\文档\Git Project> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\HP\OneDrive\文档\Git Project> git log --oneline --all --graph
* c6b2738 (HEAD -> main, origin/main) fix: correct transform test
* 16e9208 test: intentionally break transform test
* ace1428 fix: update GitHub Actions workflow
* ef23ea9 ci: add GitHub Actions workflow
* 49fe6de test: add transform function test
* 507fdc3 merge: resolve validation conflict
| \
| * 9228ef2 (feature/add-null-check) feat: add ETL completion message
| * 51003bd feat: add null validation
| * 7b6d373 feat: add processing message
| /
* 3515061 (tag: v1.0.0) feat: add load function
* 9c316e2 feat: add transform function
* 900d9a9 feat: add extract function
PS C:\Users\HP\OneDrive\文档\Git Project>
```

Figure 7 - Commit, push, and full git log graph showing the merge and CI commits

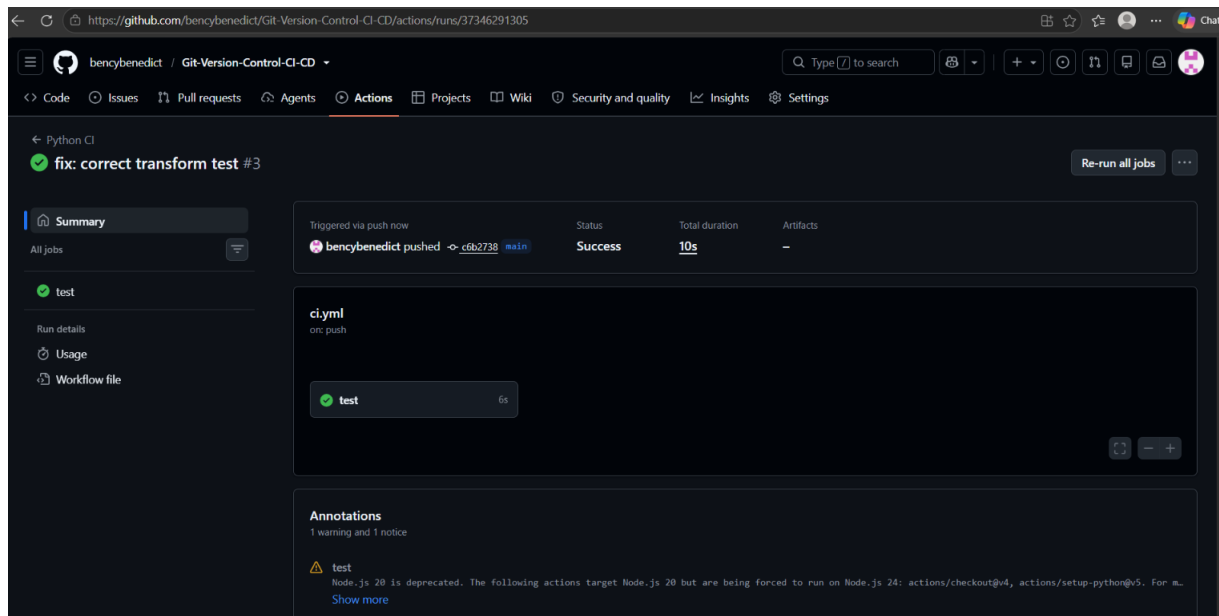


Figure 8 - GitHub Actions run #3: Success

Learned: remotes, pushing, CI feedback loop.

5. Final Repository State

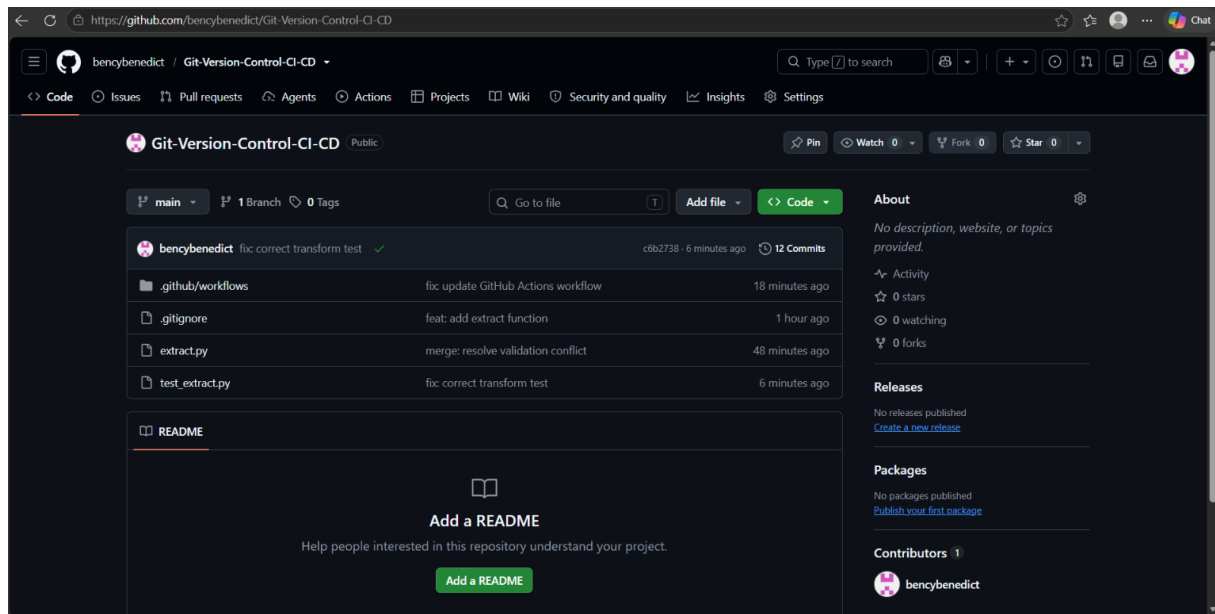


Figure 9 - GitHub repo: .github/workflows, .gitignore, extract.py, test_extract.py, 12 commits

6. Final Commit History

```
c6b2738 (HEAD -> main, origin/main) fix: correct transform test
16e9208 test: intentionally break transform test
ace1428 fix: update GitHub Actions workflow
ef23ea9 ci: add GitHub Actions workflow
49fe6de test: add transform function test
507fdc3 merge: resolve validation conflict
|\
| * 9228ef2 (feature/add-null-check) feat: add ETL completion message
| * 51003bd feat: add null validation
* | 7b6d373 feat: add processing message
|/
* 3515061 (tag: v1.0.0) feat: add load function
* 9c316e2 feat: add transform function
* 900d9a9 feat: add extract function
```